

PAPER284: M16 Eagle Nebula UQFF — Dual Mass Co-Action Product (Φ_{dm})

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Session: 80 | Module: M16UQFFMODULE.cpp (22nd C++ UQFF module)

PAPER284: M16 Eagle Nebula UQFF — Dual Mass Co-Action Product (Φ_{dm})

SFR Growth × Photoevaporation Erosion Multiplicative Coupling

Classification: UQFF 2.0 Gravitational Physics — Nebular Mass Dynamics **System:** M16 Eagle Nebula (IC 4703), Eagle Nebula Star-Forming Region **Session:** 80 | **Module:** M16UQFFMODULE.cpp (22nd C++ UQFF module) **Author:** Daniel T. Murphy | **Date:** March 2026

Abstract

This paper introduces the **Dual Mass Co-Action Product** (Φ_{dm}) — a UQFF gravity modulation factor that couples star-formation-driven mass accumulation and radiation-driven photoevaporation erosion through a **multiplicative** product rather than the previously used additive form. For M16 (Eagle Nebula), with star formation rate $SFR = 1 M_{\odot}/yr$ over initial gas mass $M_{\odot} = 1200 M_{\odot}$, and maximum photoevaporation fraction $E_{\odot} = 0.3$ (30%), the multiplicative form produces a 24.3% reduction in Φ_{dm} relative to the additive approximation at $t = 5$ Myr. This is the **first UQFF module** to simultaneously apply an additive-gain and saturation-subtractive product on the same gravity term.

2. Physical Motivation

In active star-forming nebulae, two competing processes drive mass evolution:

Star Formation Accretion — molecular gas accretes onto protostars, increasing the effective gravitational mass fraction by $SFR_{rate} \times t$.

Photoevaporation Erosion — UV radiation from newly formed massive stars erodes the surrounding gas, progressively reducing the effective mass by a saturating fraction $E_{rad}(t)$.

The **additive approximation** (used in prior UQFF modules):

$$\Phi_{dm}^{add} = g_{base} \times (1 + M_{sf}) - E_{rad}$$

linearly superposes the two effects, implicitly treating them as independent processes acting on separate mass reservoirs.

The **multiplicative form** (this paper):

$$\Phi_{dm}(t) = (1 + SFR_{rate} \times t) \times (1 - E_{rad}(t))$$

correctly encodes that **the eroded mass is drawn from the same growing reservoir** — the fraction lost to photoevaporation scales with the mass being accreted, not the original quiescent mass. This is physically accurate for pillar-geometry star formation (e.g., M16's "Pillars of Creation").

3. Mathematical Formulation

3.1 Parameters (M16 Eagle Nebula)

Parameter	Value	Description
M■	2.387×10^{33} kg (1200 M■)	Initial nebula gas mass
SFR	1 M■/yr	Star formation rate
SFR_rate	2.639×10^{-11} s ⁻¹	= SFR / (M■/M■) / (3.156×10■ s/yr)
τ_erode	9.468×10^{13} s (3 Myr)	Photoevaporation e-folding time
E■	0.3	Maximum photoevaporation fraction
g_base	1.454×10^{-12} m/s ²	$G \times M / r^2$ at $r = 3.31 \times 10^1$ ■ m

3.2 Dual Co-Action Product

M_{sf}(t) = \text{SFR_rate} \times t

E_{rad}(t) = E_0 \left(1 - e^{-t/\tau}\right)

$\boxed{\Phi_{dm}(t) = (1 + M_{sf}) \times (1 - E_{rad})}$

3.3 Dynamic Gravity Term

g_{dyn}(t) = g_{base} \times \Phi_{dm}(t)

3.4 Multiplicative–Additive Gap

The gap relative to the additive form:

$\Delta_{gap} = \Phi_{dm}^{mult} - \Phi_{dm}^{add} = -(M_{sf} \times E_{rad})$

This cross-term is always **negative** — the multiplicative form predicts lower gravity than the additive approximation whenever both SFR accumulation and erosion are simultaneously active.

4. Numerical Results at t = 5 Myr

t = 5 Myr = 1.578×10^1 ■ s

Quantity	Value
Msfrac	SFR_rate × t = $2.639 \times 10^{-11} \times 1.578 \times 10^1$ ■ = 4164.8
E_rad	E■ × (1 − exp(−5/3)) = 0.3 × 0.8110 = 0.2433
Φ_dm (multiplicative)	(1 + 4164.8) × (1 − 0.2433) = 4165.8 × 0.7567 = 3151.9
Φ_dm (additive)	(1 + 4164.8) − 0.2433 = 4165.6
gap ^{multadd}	−(4164.8 × 0.2433) = −1013.3 (24.3% less)
g_dyn(5 Myr)	$1.454 \times 10^{-12} \times 3151.9 = 4.583 \times 10^{-9}$ ■ m/s ²

The multiplicative gap of −1013.3 confirms that treating erosion as acting on the growing mass reservoir (not the static initial mass) produces a **measurable 24.3% reduction** compared to the additive approximation.

5. Connection to UQFF 2.0 g_total

In the full M16 UQFF 2.0 equation:

$$g_{\text{total}}(r, t) = \left[g_{\text{dyn}}(t) + U_{g,\text{sum}}(26) + \Lambda + Q + L + F + g_{\text{exp}} \right] \times \text{corr}_{\text{SC}}$$

The Φ_{dm} product modulates only the dynamic base gravity term g_{dyn} . The 26-layer Triadic ($U_{g,\text{sum}}$), cosmological Λ , quantum, Lorentz, fluid, and Friedmann expansion terms are all independent of Φ_{dm} — the modulation is cleanly scoped to the time-evolving mass component.

6. UQFF Historical Distinction

Module	SFR Term	Erosion Term	Form
Session 55 CP3 M16EagleNeb ulaRadiationSFR	$g_{\text{base}} \times (1 + M_{\text{sf}})$	$-E_{\text{rad}}$	Additive
This paper (PAPER_284)	$(1 + M_{\text{sf}})$	$\times (1 - E_{\text{rad}})$	Multiplicative

This is the **first UQFF module** to use the multiplicative dual co-action form, correctly encoding the coupled feedback between star formation accretion and pillar photoevaporation for M16-class nebulae.

7. Wolfram KB Term

M16UQFF:Phi_dm=(1+SFR_rate*t)*(1-E_0*(1-Exp[-t/tau])); SFR_rate=2.639e-11/s;
M_sf_frac=SFR_rate*t [PAPER_284]

8. Cross-References

- PAPER_285:** Erosion Saturation Half-Time (t_{half} , Δg_{Max})
- PAPER_286:** Nebular Friedmann Redshift (κ_{neb} , $z=0.0015$)
- M16UQFFMODULE.cpp:** Full UQFF 2.0 C++ implementation (22nd module)
- CondensedPhysics3.py:** M16DualMassCoActionProductCalculator

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